Southern New Hampshire University

CS-360: Assignment 3-2: Memorandum describing the creation of a simple Android application project

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1 Background

This will just be a simple writeup for describing the basic steps that I took to create a simple Android application project that takes in an input text through an `EditText` widget, and prints out "Hello {name}" to a `TextView` label, after clicking on, or tapping, a button. This short memorandum will document specific things that I did to create the project as specified in the rubric (CS-360, n.d.), and is written more as to give a "peep behind the scenes," in software development, as opposed to a technical description regarding the project that I created.

1.1 New Project (basic mistakes)

I began this project by first starting *Android Studio* and creating a new project. I initially created an project with the wrong settings— mainly that I selected an "Empty Activity" instead of the "Empty Views Activity," and incorrectly selected "Kotlin" as opposed to "Java." Because I could not identify the `activity_main.xml` file, I did a cursory search online as to why this might be the case, and came across a *StackOverflow thread*,[1] discussing the difference between "Empty Views" and "Empty Views Activity." I also realised that I did not properly configure what language that the app would be written in, forgetting to configure the project's language to be from Kotlin and Java.

[1] Asked by "TheMixy" (Apr. 18, 2023). *Missing activity_main.xml in new project in Android Studio*. Retrieved on Sept. 16, 2026 from: https://stackoverflow.com/questions/76048257/missing-activity-main-xml-in-new-project-in-android-studio

1.2 New Project (corrected version)

I remedied the problem described in the previous subsection by starting a new project, and paying extra attention to the settings for the new project creation. Figures 1 and 2 depicts some of the work that I did when creating a new project, going through the options and ensuring that they are in accordance with the rubic's specifications as what kind of template to use, what language to use, what minimum SDK to use, and what build configuration language to use.[2]

This creates a new *Empty Views Activity* project, with the minimum SDK version at "API 34" for Android 14.0, implemented in the Java language and build with a *Groovy DSL* build configuration manager.

1.3 Adding controls to main screen

This step was less difficult: I simply switched over onto the `activity_main.xml` codebase, and was delighted to find a "what you see is what you get" visual of the XML source code, rendered into a nice "drag and drop" interface so that I do not need to edit the XML source. The `activity_main.xml` automatically came with the default project.

I simply dragged and dropped the needed button and `EditText` text bar widgets onto the palette, and having its properties changed in accordance with CS-360 (n.d.). Figures 3 shows the `activity_main.xml` visual pane before I made my adjustments, and figure 4 depicts that very same visual pane after I have added more widgets, and made my adjustments. The raw XML source code of `activity_main.xml` can be found in the appendix for any readers interested in a

[2] See (CS-360, n.d., § entitled "Create a new Android Studio project using the Empty Views Activity template").

more precise treatment of the set up regarding the widgets.

2 End Matter

Unfortunately, there is not much that I can discuss regarding the specific steps taken to create the base project by which to develop a future "inventory app." Rather, I can only give screenshots of the *Android Studio* integrated development environment, a raw dump of the `activity_main.xml` source code, and a narriative by which to explain how I was able to create the project. The last sections and subsections accomplished the "narriative" bit, and the following sections will demonstrate screenshots and relevant portions of the project codebase in its current form.

References

CS-360 (n.d.). *Module Three Assignment Guidelines and Rubric*. Southern New Hampshire University.

3 Figures used in this paper

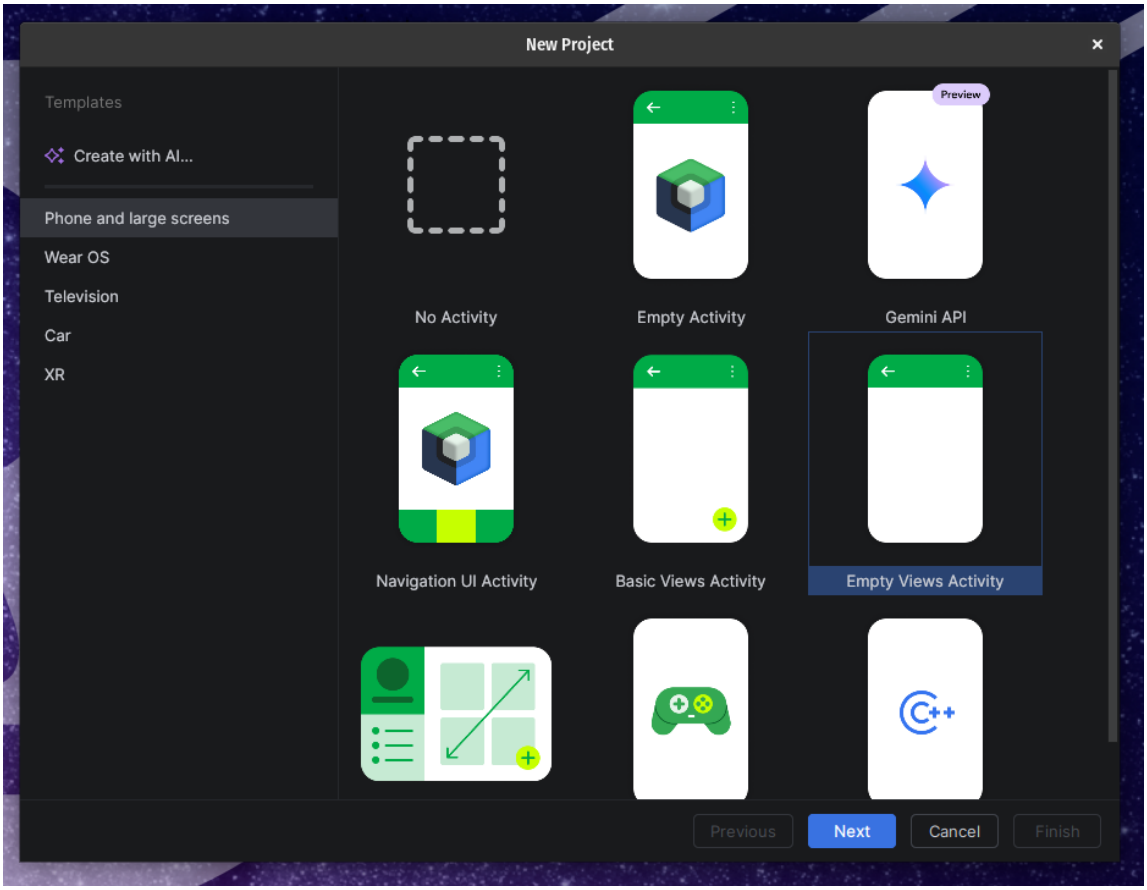


Figure 1:

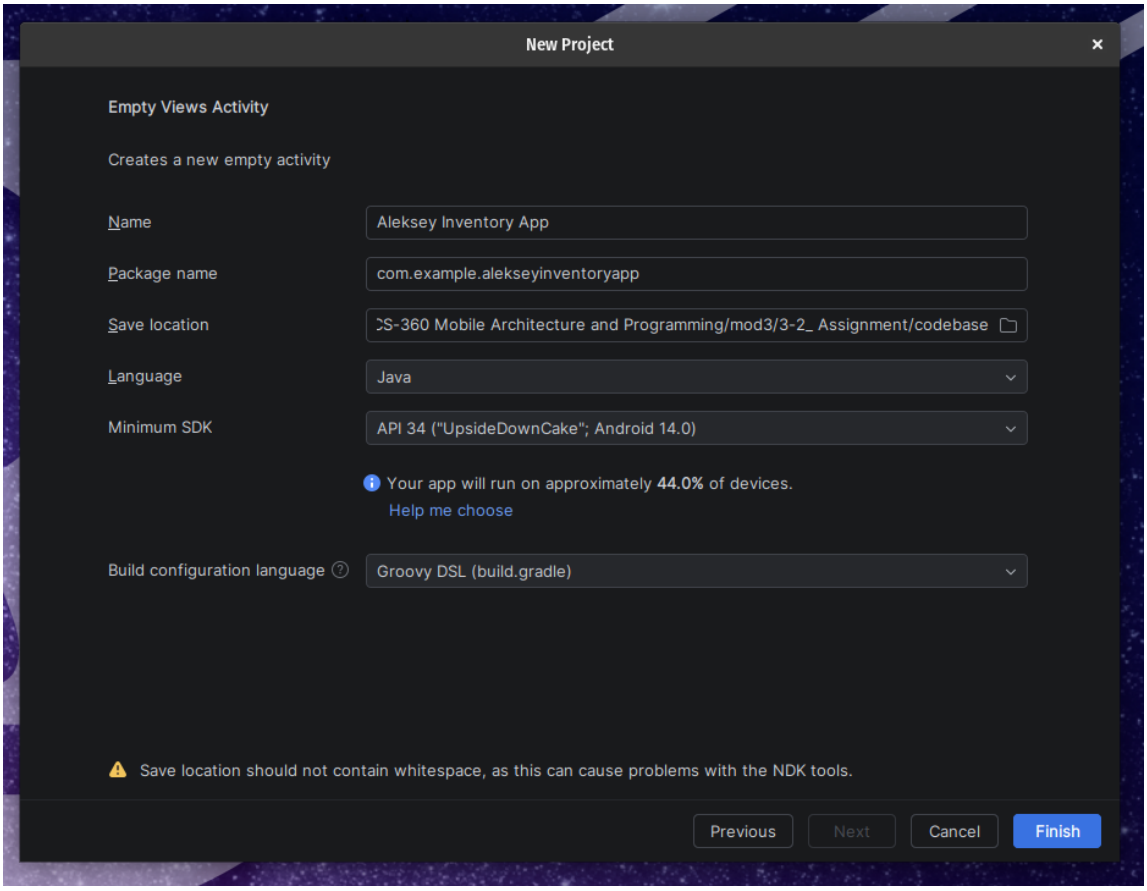

New Project
Empty Views Activity
Creates a new empty activity
Name
Aleksey Inventory App
Package name
com.example.alekseyinventoryapp
Save location
CS-360 Mobile Architecture and Programming/mod3/3-2_ Assignment/codebase
Language
Java
Minimum SDK
API 34 ("UpsideDownCake"; Android 14.0)
Your app will run on approximately 44.0% of devices.
Help me choose
Build configuration language
Groovy DSL (build.gradle)
Save location should not contain whitespace, as this can cause problems with the NDK tools.
Previous
Next
Cancel
Finish

Figure 2:

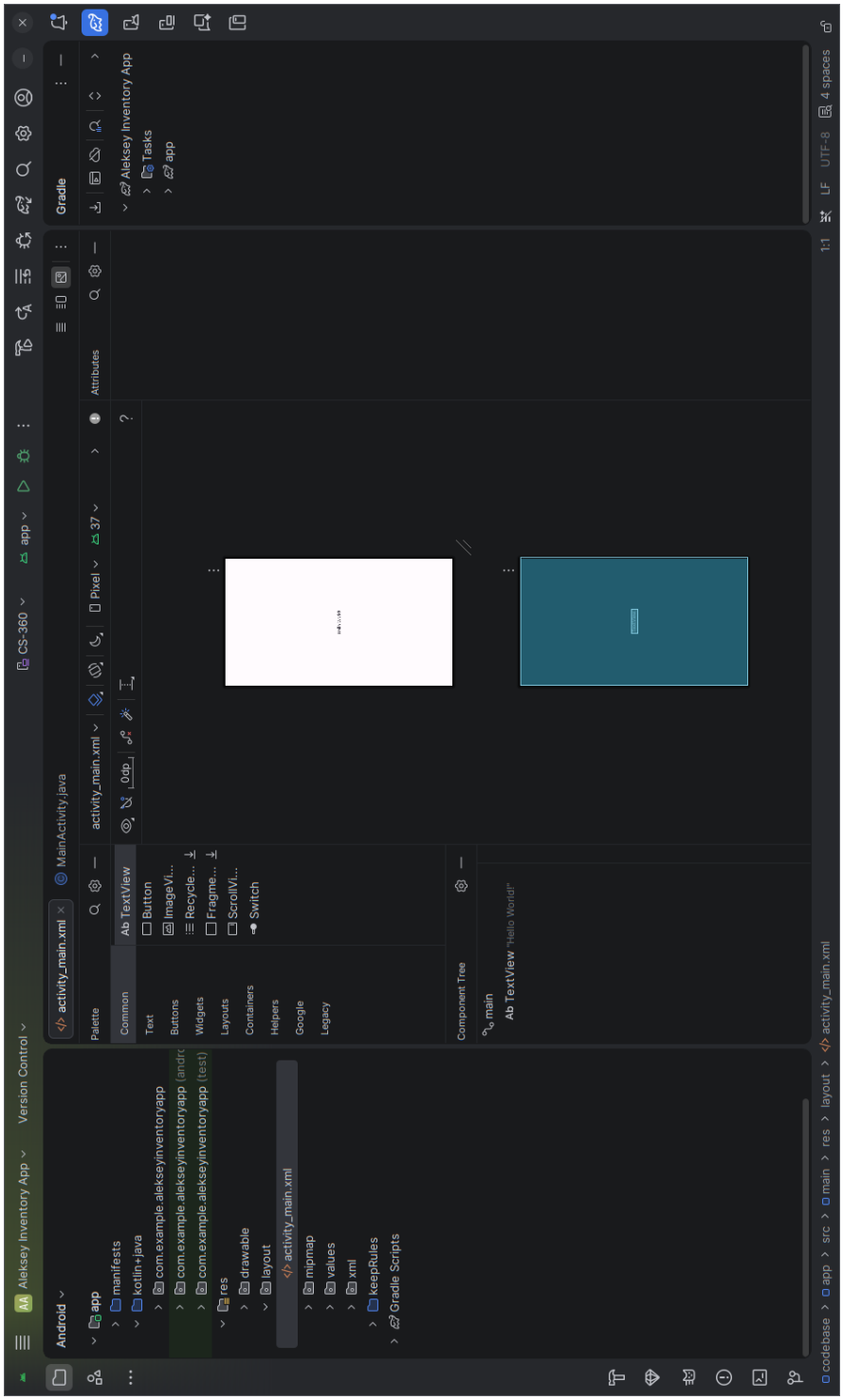

Aleksey Inventory App
Version Control
CS-360
app
Pixel
37
activity_main.xml
MainActivity.java
Palette
Attributes
Gradle
Aleksey Inventory App
Tasks
app
Android
manifests
kotlin+java
com.example.alekseyinventoryapp
com.example.alekseyinventoryapp (andro
com.example.alekseyinventoryapp (test)
res
drawable
layout
activity_main.xml
mipmap
values
xml
keepRules
Gradle Scripts
Common
Text
Buttons
Widgets
Layouts
Containers
Helpers
Google
Legacy
TextView
Button
ImageVi...
Recycle...
Fragme...
ScrollVi...
Switch
Component Tree
main
TextView "Hello World!"
codebase
app
src
main
res
layout
activity_main.xml
1:1
LF
UTF-8
4 spaces

Figure 3:

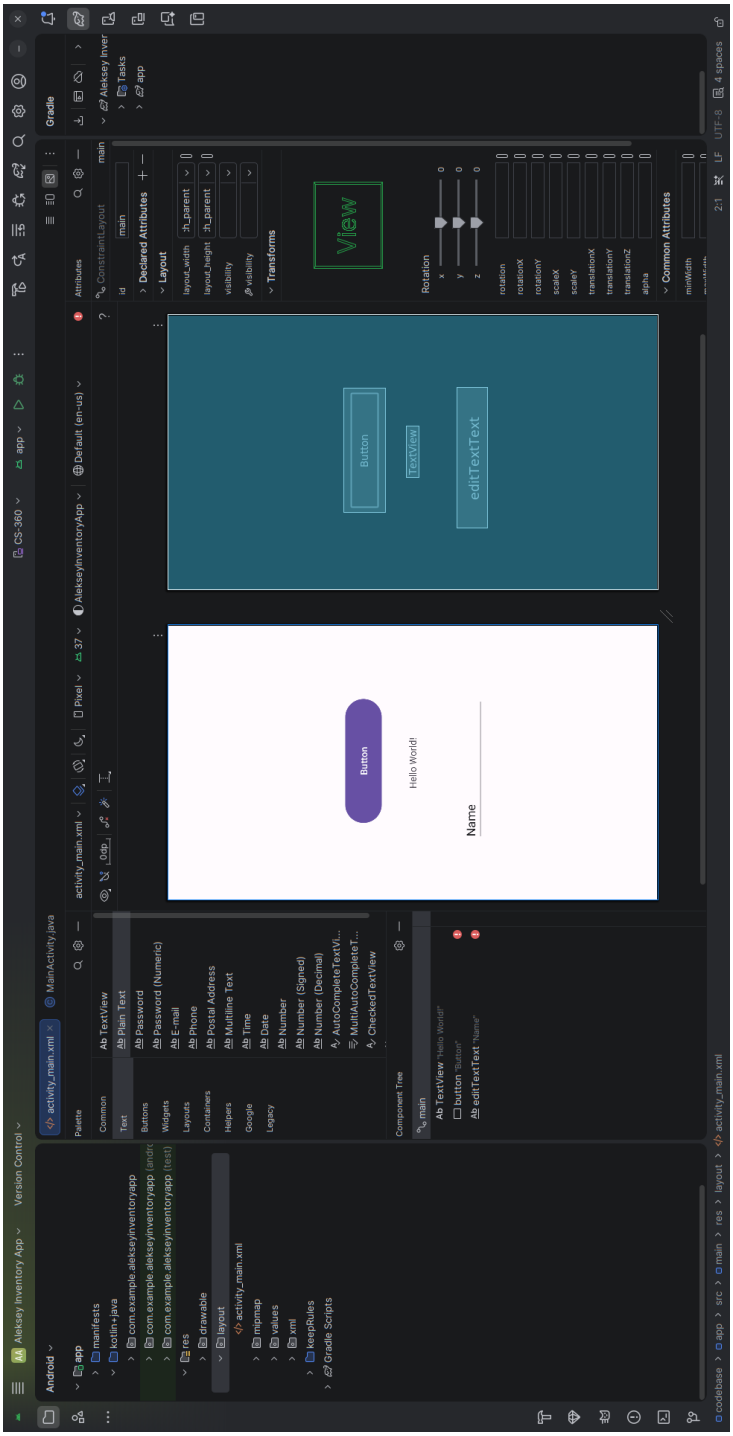


Figure 4:

4 Contents of `activity_main.xml`

```
Ln.
1. <?xml version="1.0" encoding="utf-8"?>
2. <androidx.constraintlayout.widget.ConstraintLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
3.     xmlns:app="http://schemas.android.com/apk/res-auto"
4.     xmlns:tools="http://schemas.android.com/tools"
5.     android:id="@+id/main"
6.     android:layout_width="match_parent"
7.     android:layout_height="match_parent"
8.     tools:context=".MainActivity">
9.
10.     <TextView
11.         android:id="@+id/textGreeting"
12.         android:layout_width="wrap_content"
13.         android:layout_height="wrap_content"
14.         app:layout_constraintBottom_toBottomOf="parent"
15.         app:layout_constraintEnd_toEndOf="parent"
16.         app:layout_constraintStart_toStartOf="parent"
17.         app:layout_constraintTop_toTopOf="parent" />
18.
19.     <Button
20.         android:id="@+id/buttonSayHello"
21.         android:layout_width="186dp"
```

```
22.         android:layout_height="62dp"
23.         android:text="Say Hello"
24.         tools:layout_editor_absoluteX="116dp"
25.         tools:layout_editor_absoluteY="262dp" />
26.
27.     <EditText
28.         android:id="@+id/nameText"
29.         android:layout_width="wrap_content"
30.         android:layout_height="wrap_content"
31.         android:ems="10"
32.         android:hint="Enter your name"
33.         android:inputType="text"
34.         tools:layout_editor_absoluteX="92dp"
35.         tools:layout_editor_absoluteY="431dp" />
36. </androidx.constraintlayout.widget.ConstraintLayout>
```